

BayWatch: REST API for Key West Coastal Analysis

For my final project, I want to build a REST API around coastal water level data from the NOAA's Center for Operational Oceanographic Products and Services, using the Key West and Lake Michigan stations as the main datasets. This data has time-stamped records and is good for both CRUD and data analysis. There are over 200 water level stations on the coasts and Great Lakes, however, I want to focus on Key West because of the outliers from the constant hurricanes, and then Lake Michigan to compare with more steady-level data.

This data source is especially good for this project because NOAA provides a public API for getting the data in a structured format, such as JSON, which I'm already comfortable with from the previous assignment. There are both 6-minute and hour interval datasets, and I think I will be focusing on the hourly datasets because it will show more movement, as well as be easier to read through.

Each reading from the hour intervals can be represented as a Pydantic object with fields such as station ID, station name, timestamp, water level, high/low, and daily means (I'm attaching a JSON example below). This structure is great for Pydantic and will make it simple to build summary statistics and filter by a time window.

From an engineering perspective, this project is important because coastal water levels directly impact the real world because of flood risk and navigation planning. I was debating between using the Lake Michigan station because it is closer to home, and I've actually been there. I decided to do both because the Key West data is likely more dynamic, which will lead to more interesting and summary statistics that are also practically relevant. And then the Lake Michigan data should be more consistent day-to-day, providing a good comparison for the variability in Key West, as well as increasing the overall number of routes in my model.

I plan to build an API that allows users to load these water-level readings into Redis, view all observations and certain groups, retrieve specific readings, and delete specific or all records. Users will also be able to send analysis jobs that will be handled by workers. An example could be comparing the water levels between the two sites or summarizing the highest and lowest observations in a selected time range.

I'm excited about the topic, so I think it will be manageable for a solo project.

Data Format: <https://api.tidesandcurrents.noaa.gov/api/prod/>

Data link for the JSON version:

https://api.tidesandcurrents.noaa.gov/api/prod/datagetter?product=hourly_height&application=Ti deTrack&begin_date=20250101&end_date=20250107&datum=MLLW&station=8724580&time_zone=gmt&units=metric&format=json

```
{  
  "station_id": "8724580",  
  "station_name": "Key West",  
  "timestamp": "2025-01-01 00:00",  
  "water_level": 0.593,  
  "sigma": 0.005,  
  "daily_high": null,  
  "daily_low": null,  
  "daily_mean": null  
}
```